Scepter C™ / Scepter XC™ Occlusion Balloon Catheter

Instructions for Use

DEVICE DESCRIPTION

The Scepter C and Scepter XC Occlusion Balloon Catheters are dual lumen catheters with an external hydrophilic coating. The guidewire lumen is provided for guidewire introduction and delivery of agents. The inflation lumen is used exclusively for the inflation and deflation of the balloon. The guidewire lumen of the balloon catheters are compatible with 0.014 inch or smaller guidewires and the balloon can be inflated and deflated independently with or without the presence of a guidewire. The balloon catheters are available in different sizes and lengths and incorporate radiopaque markers to facilitate fluoroscopic visualization and indication of the balloon position. The balloon incorporates a distal air-purge hole to purge air from the inflation lumen prior to use.

The Scepter C and Scepter XC Occlusion Balloon Catheters are intended for single use only. Do not resterilize or reuse the balloon catheters. After use, dispose of the catheter in accordance with hospital, administrative and/or local government policy. Do not use the balloon catheters if the sterile package is breached or damaged.

CONTENTS

One Occlusion Balloon Catheter
One Introducer Sheath
One Shaping Mandrel
One Compliance Card

INDICATIONS FOR USE / INTENDED PURPOSE

The Scepter C and Scepter XC Occlusion Balloon Catheters are intended:

For use in the peripheral and neuro vasculature where temporary occlusion is desired. The balloon catheters provide temporary vascular occlusion which is useful in selectively stopping or controlling blood flow. The balloon catheters also offer balloon assisted embolization of intracranial aneurysms.

For use in the peripheral vasculature for the infusion of diagnostic agents, such as contrast media, and therapeutic agents such as embolization materials.

For neurovascular use for the infusion of diagnostic agents such as contrast media, and therapeutic agents, such as embolization materials, that have been approved or cleared for use in the neurovasculature and are compatible with the inner diameter of the Scepter C/XC Balloon Catheter.

CONTRAINDICATIONS

Not intended for embolectomy or angioplasty procedures
Not intended for use in coronary vessels
Not intended for pediatric or neonatal use

CAUTION

Rx Only: Federal (USA) law restricts this device to sale by or on the order of a physician.

Do not use if pouch is opened or damaged.

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This device is intended for single use only. Do not resterilize or reuse. Reuse, reprocessing or resterilization may compromise the structural integrity of the device and/or lead to device failure which, in turn, may result in patient injury, illness, or death. Reuse, reprocessing, or resterilization may also create a risk of contamination of the device and/or cause patient infection or cross-infection, including, but not limited to, the transmission of infectious disease(s) from one patient to another. Contamination of the device may lead to injury, illness or death of the patient.

After use, dispose in accordance with hospital, administrative and/or local government policy.

WARNINGS

Verify the size of the vessel under fluoroscopy. Ensure that the balloon catheter is appropriate for the size of the vessel.

Do not exceed the maximum recommended inflation volume as balloon rupture may occur.
The balloon catheter has been tested for compatibility or use with Onyx™ Liquid Embolic System and DMSO. For all other liquid embolics, refer to their Instructions For Use.
The balloon catheter is provided sterile and non-pyrogenic. Do not use if the packaging is breached or damaged.

Viscosity and concentration of contrast will affect balloon inflation and deflation times.

During preparation, do not deflate the balloon unless the distal tip is submerged in saline or contrast to prevent air from entering balloon.

Do not attach any high pressure devices to the balloon inflation port as this may rupture the balloon.

Do not inflate the balloon with air or any other gas while in the body.

Improper preparation may introduce air into the system. The presence of air may inhibit proper fluoroscopic visualization.

Excessive pressure higher than 700 PSI (4826kPa, 47.6atm) may cause leakage or rupture of the balloon catheter guidewire lumen.
When air-purging the balloon catheter, inject fluid slowly otherwise balloon rupture may occur.

If back-loading the balloon catheter over a guidewire, ensure distal tip of the balloon catheter is not damaged.

Do not over-tighten the RHV around the balloon catheter. Over-tightening could delay balloon inflation and deflation.

Do not advance the balloon catheter or guidewire against resistance. If resistance is felt, assess the source of resistance using fluoroscopic means.

Always inflate and deflate the balloon while visualizing under fluoroscopy to ensure patient safety.

The shaping mandrel is not intended for use inside the body. Ensure shaping mandrel is removed from balloon catheter prior to introduction into the RHV or other accessories.

NBCA and solutions containing ethyl esters of iodized fatty acids of poppy seed oil are not compatible with the balloon.

PRECAUTIONS

Immediately prior to use visually inspect all the sterile barrier systems, that are labeled as sterile. Do not use if breaches in sterile barrier system integrity are evident such as a damaged pouch.

After balloon preparation for use and prior to use, re-inflate to nominal volume and inspect for any irregularities or damage. Do not use if any inconsistencies are observed.

Verify balloon catheter compatibility when using other ancillary devices commonly used in intravascular procedures. Physician must be familiar with percutaneous, intravascular techniques and possible complications associated with the procedure.

The balloon catheter has a lubricious surface and should be hydrated prior to use. Once the balloon catheter is hydrated, do not allow it to dry.

Protect the balloon when tip steam shaping or purge hole sealing of the balloon catheter as it may affect the integrity of the balloon material.

Exercise care in handling the balloon catheter to reduce the chance of accidental damage.
With the exception of dimethyl sulfoxide (DMSO), use of other organic solvents may damage the balloon catheter and/or coating on the surface.

Verify that the diameter of any guidewire or accessory device used is compatible with the inner diameter of the balloon catheter prior to use.

Take precaution when manipulating the balloon catheter in tortuous vasculature to avoid damage. Avoid advancement or withdrawal against resistance until the cause of resistance is determined.

Presence of calcifications, irregularities or existing devices may damage the balloon catheter and potentially affect its insertion or removal.

Always verify proper balloon vessel occlusion prior to and during embolic material delivery. Sealing the purge hole prior to embolic material delivery may provide assistance during use.

Excessive torque applied to the syringe might result in damage to the Scepter hub assembly.

Continuing negative aspiration after the balloon is fully deflated will result in blood entering the balloon and will reduce fluoroscopic visibility.

POTENTIAL COMPLICATIONS

Potential complications include but are not limited to: vessel or aneurysm perforation, vasospasm, hematoma at the site of entry, embolism, ischemia, intracerebral/intracranial hemorrhage, pseudoaneurysm, seizure, stroke, infection, vessel dissection, thrombus formation, and death.

COMPATIBILITY

Scepter C and Scepter XC balloon catheters are compatible with .014” (0.36mm) or smaller guidewires
Note: Guidewire not required for inflation of balloon

Choose appropriate guiding catheter with minimum inner diameter larger than or equal to 0.053” (1.35mm)
Note: The maximum outer diameter of the balloon catheter is 0.038” (0.97mm)
Scepter C and Scepter XC balloon catheters are compatible for use with dimethyl sulfoxide (DMSO).
Scepter C and Scepter XC balloon catheters have been verified to be compatible for use with diagnostic agents (such as contrast media) and liquid embolic agents (i.e. Onyx™ Liquid Embolic System). For all other liquid embolics, refer to their Instructions For Use.

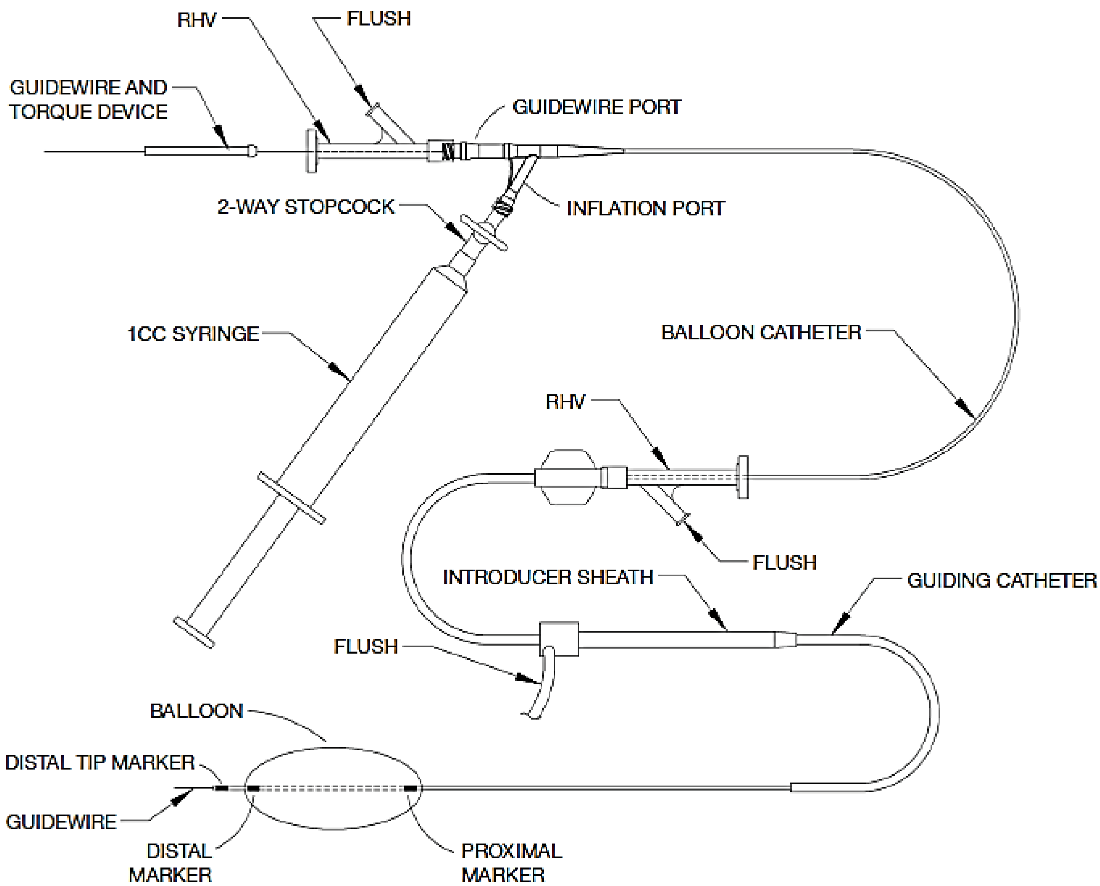

DIAGRAM OF BALLOON CATHETER SETUP

HYDRATION FLUSH

1. Choose a balloon catheter that is appropriate for the size of the vessel.

2. Before removing the balloon catheter from the dispenser tube, gently pull the hub out from the dispenser tube in order to loosen the strain relief from the dispenser tube, fully hydrate the hydrophilic segment of the device by flushing heparinized saline through the dispenser tube using a syringe attached to the flush port. Allow 30 seconds of hydration time.

Table 1: Approximate Balloon Deflation Time

Contrast Name	Viscosity @ 37˚C (cps)	Contrast:Saline	**Scepter C (seconds)**			**Scepter XC (seconds)**
			4x10mm	4x15mm	4x20mm	4x11mm
Omnipaque 300	6.3	50:50	≤ 15	≤ 18	≤ 20	≤ 14

Table 2:

Approximate Prime Volume of the Entire Inflation Lumen	**Approximate Prime Volume of the Entire Guidewire Lumen**
Volume of Inflation Lumen + Inflation Hub	Volume of Guidewire Lumen + Guidewire Hub

0.45cc	**0.44cc**

BALLOON PREPARATION

1. Use a syringe with heparinized saline to flush the guidewire lumen. Remove the syringe. Carefully introduce a hydrated guidewire into the guidewire lumen of the balloon catheter.
   WARNING: Excessive pressure higher than 700PSI (4826kPa, 47.6atm) may cause leakage or rupture of the balloon catheter.

2. Remove the balloon catheter by pulling it from the dispenser tube. If resistance is observed, repeat the flushing procedure in preparation for use until the balloon catheter is well hydrated and can be easily removed from the dispenser tube. Inspect the balloon catheter thoroughly to ensure it is not damaged. Do not allow balloon catheter to dry prior to introduction into the guiding catheter. Do not reinsert a hydrated balloon catheter into its packaging.

3. Prepare a contrast/saline solution using Table 1 as a guide.
   WARNING: Viscosity and concentration of contrast will affect balloon inflation and deflation times.

4. Fill a 10cc syringe with contrast/saline solution and carefully attach directly to inflation port without injecting contrast into hub. Ensure there are no bubble(s) in syringe prior to attaching

5. Hold balloon proximal to inflation plug and point balloon upright with one hand.

6. Hold the attached syringe upright (pointing up) with the other hand and apply pressure on syringe plunger using thumb.

7. If the balloon is initially inflated with air then maintain constant syringe pressure.
   WARNING: Do not fully inflate balloon with air. Maintain balloon inflation below distal balloon marker band.

8. Maintain pressure and DO NOT TILT the balloon until the contrast reaches the distal purge hole and the contrast has completely filled the balloon.

9. Once the balloon has been fully inflated with contrast, inspect balloon for any damages and bubbles. If air bubbles persist, use the aspiration technique outlined in Step A otherwise place tip in saline bowl, deflate balloon.

10. Remove the 10cc syringe and attach a stopcock to a 1cc syringe filled with contrast/saline solution.

11. Prime the 1cc syringe and stopcock with contrast/saline solution, attach to the hub of the primed inflation port and proceed to step B, C or D.

ALTERNATE BALLOON PREPARATION

1. Remove the balloon catheter by pulling it from the dispenser tube. If resistance is observed, repeat the flushing procedure until the balloon catheter is well hydrated and can be easily removed from the dispenser tube. Inspect the balloon catheter thoroughly to ensure it is not damaged. Do not allow balloon catheter to dry prior to introduction into the guiding catheter. Do not reinsert a hydrated balloon catheter into its packaging.

2. Hydrate the guidewire lumen of the balloon catheter with saline. See table 2 for approximate prime volume. Carefully introduce a hydrated guidewire into the guidewire lumen of the balloon catheter.

WARNING: Excessive pressure higher than 700PSI (4826kPa, 47.6atm) may cause leakage or rupture of the balloon catheter.
WARNING: If back-loading the balloon catheter over a guidewire, ensure distal tip of the balloon catheter is not damaged.

3. Prepare a contrast/saline solution using Table 1 as a guide.
   WARNING: Viscosity and concentration of contrast will affect balloon inflation and deflation times.

4. Fill the hub of the inflation port with contrast/saline solution using an introducer needle attached to a 1cc syringe.

5. Remove the introducer needle and attach a stopcock to a 1cc syringe filled with contrast/saline solution.

6. Prime the stopcock with contrast/saline solution and attach it to the hub of the primed inflation port.

7. Place balloon catheter into a large bowl of saline for continued hydration. Note: keep the catheter in a horizontal position while submerged in the saline solution. With the distal tip of the balloon catheter submerged in saline, slowly inject contrast/saline solution through the balloon inflation port and allow all the air to escape from the distal air-purge hole. During this air-purging process, caution should be taken to inject slowly to prevent the balloon from inflating with the air. If the balloon is inflated before the contrast/saline solution reaches the balloon, stop air-purging the balloon to allow the balloon to deflate. Slowly inject contrast/saline solution again allowing remaining air to escape through the distal air-purge hole. Reference table 2 for the total prime volume of the inflation hub and the inflation lumen. Additional solution will be required to inflate the balloon.
   WARNING: During air-purging of balloon catheter, inject fluid slowly otherwise balloon rupture may occur.

8. After air-purging the balloon catheter inflation lumen, continue to slowly inflate the balloon with the contrast/saline solution. The balloon will inflate from proximal to distal during the initial inflation. When the balloon catheter is completely inflated, check for the presence of additional air bubbles. If any air bubbles are present, submerge the tip of the balloon catheter in saline and deflate completely, then slowly inflate with the balloon held upright to allow any remaining air to escape from the distal air-purge hole. If bubbles persist, use the aspiration technique outlined in Step A, otherwise place tip in saline bowl, deflate balloon and proceed to step B, C or D.
   NOTE: When air-purging, keep balloon catheter hydrated during preparation procedure to prevent coating from drying.
   WARNING: When air-purging of balloon catheter, inject fluid slowly otherwise balloon rupture may occur.
   WARNING: Do not deflate the balloon unless the distal tip is submerged in saline or contrast to prevent air from getting back into balloon.
   WARNING: Keep balloon catheter hydrated during preparation procedure by periodically submerging in saline as required.

A. ASPIRATION TECHNIQUE:
   1. Attach a 3-way stopcock to a 20cc syringe filled with 3cc of contrast/saline solution.
   2. Connect the 20cc syringe/stopcock to balloon catheter inflation port.
   3. With syringe pointed down and stopcock in open position, pull vacuum on syringe.
   4. Remove the balloon from the saline bowl and place the distal tip of the balloon in air.
   5. Hold vacuum until the inflation port is clear of contrast/saline mix.
   6. Close stopcock to inflation lumen, point syringe up and purge air.
   7. Open stopcock to inflation lumen, point syringe down and pull vacuum again.
   8. Slowly remove vacuum by lowering plunger and let the system equalize.
   9. Replace 20cc syringe with 1cc syringe filled with contrast/saline solution.
      WARNING: Keep balloon catheter hydrated during preparation procedure by periodically submerging in saline as required.
   10. Advance guidewire distally or insert a straight shaping mandrel to ensure balloon is straight.

11. Hold tip upright to allow air to escape and slowly infuse contrast/saline until balloon is inflated to nominal diameter.
12. If no air bubbles are visible inside balloon, place tip in saline bowl, deflate balloon and then proceed to Step B, C or D. If air bubbles persist, repeat step A or discard balloon catheter.

B. TIP SHAPING TECHNIQUE:

NOTE: Tip shaping must be performed **AFTER** balloon catheter has been purged and has been properly prepared per instructions.

1. Bend the shaping mandrel for desired shape.
2. With the balloon deflated, carefully insert the shaping mandrel into distal end of the guidewire lumen.
3. Hold the bottom of the compliance card with “balloon side” facing user.
4. Hold the balloon proximal to the inflation port with hand resting on center of compliance card for stability.
5. Place the distal tip marker band with bend at the bottom and inside of the steam groove. (see Figure 2)
   NOTE: Exposing more than 3mm of the distal tip to steam will result in balloon damage.
   PRECAUTION: Discretion should be taken not to contaminate the balloon catheter during steaming.

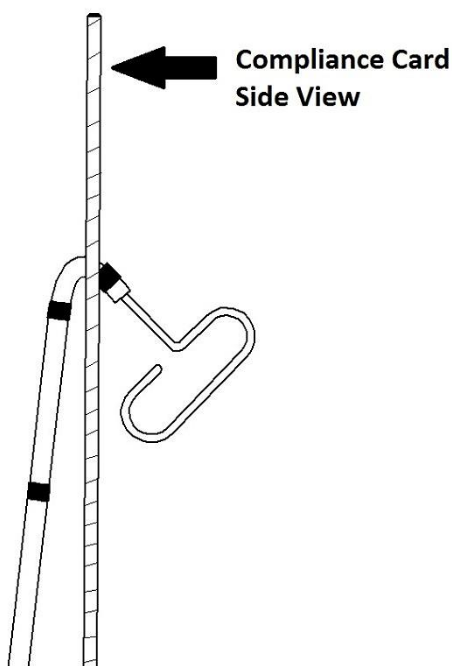

Figure 1 Shaped Configuration

6. Hold the distal tip marker band in laminar steam flow close to the source of steam for approximately 20 seconds.
   NOTE: Ensure that the compliance card is protecting the balloon from steam heat source or balloon may be damaged.
7. Remove the catheter tip and compliance card from steam.
8. Gently remove the balloon catheter tip from compliance card and quench in heparinized saline.
9. Gently remove tip shaping mandrel.
   WARNING: The shaping mandrel is not intended for use inside the body.
10. Proceed to step D.

C. AIR PURGE HOLE SEALING TECHNIQUE:

NOTE: Air-Purge hole sealing must be performed **AFTER** balloon catheter has been purged and has been properly prepared per instructions.

1. With the balloon deflated, carefully insert the shaping mandrel into distal end of the guidewire lumen.
2. Hold the bottom of the compliance card with “balloon side” facing user.
3. Hold the balloon proximal to the inflation port with hand resting on center of compliance card for stability.
4. Place the distal tip marker band at the bottom and inside of the steam groove. (see Figure 1)
   NOTE: Exposing more than 3mm will result in balloon damage.
   PRECAUTION: Discretion should be taken not to contaminate the balloon catheter during steaming.

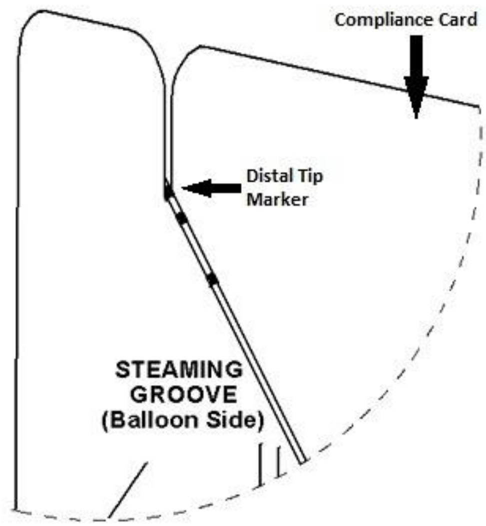

Figure 2 Straight Configuration

5. Hold the distal tip marker band in laminar steam flow close to the source of steam for approximately 20 seconds.
   NOTE: Ensure that the compliance card is protecting the balloon from steam heat source or balloon may be damaged.
6. Remove catheter tip and compliance card from steam source.
7. Gently remove the balloon catheter tip from compliance card and quench in heparinized saline. Gently remove tip shaping mandrel. **WARNING:** Shaping mandrel is not intended for use inside the body.
   NOTE: Improper exposure to steam may result in partially sealed air-purge hole. Inspection of inflated balloon catheter under fluoroscopy must be performed frequently to ensure desired occlusion at all times.
8. Proceed to step D.

D. BALLOON FINAL INSPECTION

1. Re-inflate the balloon to nominal volume to inspect the balloon catheter prior to use for any irregularities or damage. Do not use if any inconsistencies are observed.
2. If air purge hole sealing was performed, inspect the balloon catheter distal tip for any contrast leakage from air purge hole. If contrast leakage is observed then repeat air-purge hole sealing technique outlined in step C.
3. Deflate once more while distal tip is submerged in saline and let the pressure within the catheter equalize. With the catheter and balloon completely primed, the balloon catheter is ready for use.
   WARNING: Do not attach any high pressure devices to the balloon inflation port as this may rupture the balloon.

WARNING: Do not inflate the balloon with air or any other gas while in the body.
WARNING: Improper preparation may introduce air into the system. This may inhibit proper fluoroscopic visualization.

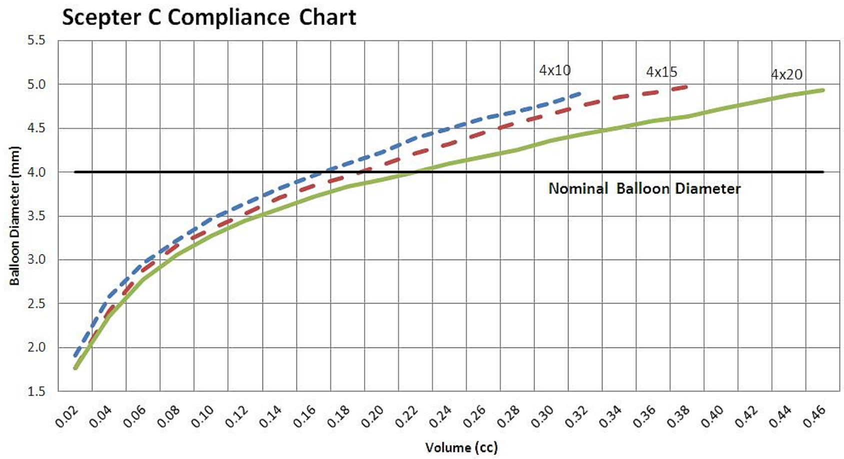

Table 3: Balloon Inflation Compliance				
Inflation Volume* (cc)	Scepter C Dia., (mm)			Scepter XC Dia., (mm)
	4x10	4x15	4x20	4x11
0.02	1.9	1.8	1.8	1.8
0.04	2.6	2.4	2.4	2.6
0.06	3.0	2.9	2.8	3.2
0.08	3.2	3.2	3.1	3.6
0.10	3.5	3.4	3.3	3.9
0.12	3.6	3.5	3.4	4.2
0.14	3.8	3.7	3.6	4.5
0.16	4.0	3.8	3.7	4.7
0.18	4.1	4.0	3.8	4.9
0.20	4.2	4.1	3.9	5.0
0.22	4.4	4.2	4.0	5.2
0.24	4.5	4.3	4.1	5.3
0.26	4.6	4.5	4.2	5.5
0.28	4.7	4.6	4.3	5.6
0.30	4.8	4.7	4.4	5.7
0.32**	4.9	4.8	4.4	5.9
0.34		4.9	4.5	
0.36		4.9	4.6	
0.38**		5.0	4.6	
0.40			4.7	
0.42			4.8	
0.44			4.9	
0.46**			4.9	

*After Priming Catheter **Maximum Injection Volume

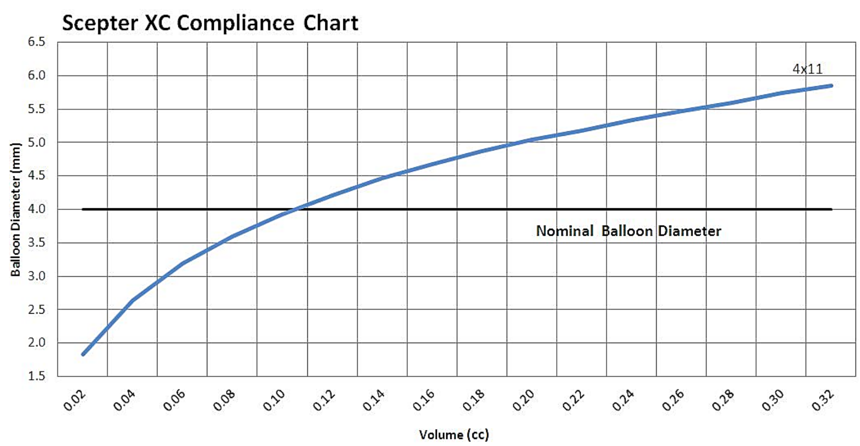

Table 4	Approximate Nominal Flow Rates at 100 and 300 psi Infusion Pressure							
	Saline		**50/50% Contrast (300mgI/ml)**		**100% Contrast (300mgI/ml)**		**100% Contrast (350mgI/ml)**	
Scepter C & XC	100 psi	300 psi	100 psi	300 psi	100 psi	300 psi	100 psi	300 psi
	0.57 cc/sec	1.50 cc/sec	0.49 cc/sec	0.71 cc/sec	0.33 cc/sec	0.62 cc/sec	0.40 cc/sec	0.40 cc/sec

DIRECTIONS FOR USE (Refer to diagram for reference)

1. Attach a rotating hemostatic valve (RHV) to the guidewire lumen of the balloon catheter. Set up a continuous saline flush line and connect it to the sidearm of the RHV.
2. Choose appropriate guiding or diagnostic catheter. Attach a RHV to the proximal hub of the guiding or diagnostic catheter. To prevent backflow of blood into the lumen of the catheter, connect the continuous saline flush line to the sidearm of the RHV.
3. Open the RHV on the hub of the guiding or diagnostic catheter and introduce the balloon catheter/guidewire into the guiding catheter using the introducer sheath. Carefully advance the balloon catheter/guidewire to the guiding catheter distal tip. After the balloon catheter/guidewire reaches the tip of the guiding catheter, remove the introducer from the balloon catheter shaft by retracting the introducer from the RHV and peeling off the introducer. Advance the balloon catheter through the RHV.
4. Advance the balloon catheter and guidewire to the desired location in the vasculature using fluoroscopic visualization. Carefully tighten the valve of the RHV around the balloon catheter to prevent leakage from the RHV. The RHV should still allow for balloon catheter advancement after tightening.
   WARNING: Do not over-tighten the RHV around the balloon catheter. Over-tightening could delay balloon inflation and deflation.
   WARNING: Do not advance the balloon catheter or guidewire against resistance. If resistance is felt, assess the source of resistance using fluoroscopic means.
5. Attach a 2-way stopcock to a 1cc syringe filled with appropriate contrast solution. Prime the 2-way stopcock so that no air is present. Slowly inflate the balloon to the recommended volume to achieve the desired diameter as described in Table 3.
   WARNING: Do not exceed the maximum recommended inflation volume as balloon rupture may occur.
   WARNING: Always inflate and deflate the balloon while visualizing under fluoroscopy to ensure patient safety.
6. After inflation, lock the stopcock if desired.
7. If desired, remove the guidewire from the balloon catheter and prepare per the respective diagnostic or therapeutic agent IFU(s) for delivery through the guidewire lumen.
   WARNING: Excessive pressure higher than 700PSI (4826kPa, 47.6atm) may cause leakage or rupture of the guidewire lumen.
8. When deflating balloon, use fluoroscopy to ensure complete deflation prior to removal. See Table 1 for respective deflation times. After procedure is complete, slowly remove the balloon catheter and guidewire.

STORAGE

Avoid exposure to water, sunlight, extreme temperatures and high humidity during storage. Store the balloon catheter under controlled room temperature. See the product label for the device shelf life. Do not use the device beyond the labeled shelf life.

MATERIALS

The balloon catheter does not contain latex or PVC materials.

SYMBOLS

LOT	Lot Number		Do Not Reuse
REF	Catalog Number		Caution
CONTENTS	Contents		Use-by Date
STERILE EO	Sterilized Using Ethylene Oxide		Date of Manufacture
CE	CE Mark		Manufacturer
EC REP	Authorized European Representative		Non-Pyrogenic
	Consult Instructions for use	Rx ONLY	For Prescription Use Only
	Single Sterile barrier system with protective packaging inside	MD	Medical Device
	Do not use if package is damaged		Country of manufacture
	Keep Dry		Keep Away from Sunlight
	Do Not Resterilize		Importer

WARRANTY

MicroVention, Inc. warrants that reasonable care has been used in the design and manufacture of this device. This warranty is in lieu of and excludes all other warranties not expressly set forth herein, whether expressed or implied by operation of law or otherwise, including, but not limited to, any implied warranties of merchantability or fitness. Handling, storage, cleaning and sterilization of the device as well as factors relating to the patient, diagnosis, treatment, surgical procedure and other matters beyond MicroVention's control directly affect the device and the results obtained from its use. MicroVention's obligation under this warranty is limited to the repair or replacement of this device and MicroVention shall not be liable for any incidental or consequential loss, damage or expense directly or indirectly arising from the use of this device. MicroVention neither assumes, nor authorizes any other person to assume for it, any other or additional liability or responsibility in connection with this device. MicroVention assumes no liability with respect to devices reused, reprocessed or resterilized and makes no warranties, expressed or implied, including, but not limited to, merchantability or fitness for intended use, with respect to such device.

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eIFU website: www.microvention/eIFU-MicroVention.com

IFU100302X1

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